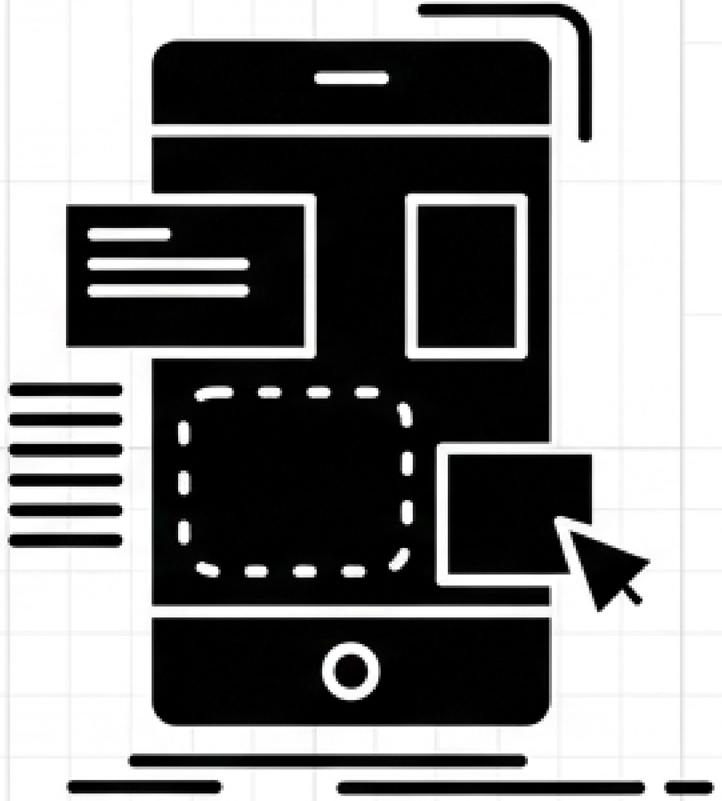
DailyKhata: AI-Powered Expense Tracking for Pakistan

Human-Centered Design Capstone Case Study



BNU

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Course: HCI & Computer Graphics (Section C)
Focus: Milestone 2 (Needs Finding) & Milestone 3 (Prototyping & Testing)

The Needs Finding Methodology

Study Goals

- Explore how users track daily expenses
- Pinpoint friction causing abandonment
- Understand mental models for categorizing physical receipts
- Gauge attitudes toward AI/voice input

1. **Intro & Consent** — 10-20 min total session, voluntary, audio-recorded.
2. **Warm-up** — Typical week & current tools.
3. **The Shopping Trip** — Walkthrough of last grocery run and receipt handling.
4. **Workflow Observation** — Show us your system.
5. **Friction & Abandonment** — Unlogged cash, app quit-rates.
6. **Concept Intro** — Gauging reactions to camera & voice AI.

Study Participants: The Dual-User Challenge

Group A (Homemakers & Local Shoppers)	Group B (Students & Freelancers)
Profile: 5 Participants (Ages 28-50). Low-to-moderate digital literacy. Primary household budget managers. **Current Tools:** Physical notebook (Khata), WhatsApp notes, mental tracking, paper ledgers.	**Profile:** 5 Participants (Ages 21-30). High digital literacy. Require granular financial oversight. **Current Tools:** Abandoned apps (Cashew, Wallet), spreadsheets, Notion.
P1: Physical notebook (Khata)	P6: Abandoned Wallet -> Spreadsheet
P2: Notebook + WhatsApp notes	P7: No consistent system
P3: Mental tracking + receipts in drawer	P8: Notion + manual receipt photos
P4: Shopkeeper / Paper ledger	P9: WhatsApp messages to self
P5: Notes app + notebook	P10: Abandoned Cashew -> mental tracking

Affinity Diagramming: 5 Core Themes from 180 Insights

Receipt Chaos

[38 notes]

"I just put it in my purse. By the time I get home, half of them are crumpled." – P2

Informal Cash Gaps

[31 notes]

"Rickshaw has no receipt, vendor just says the amount... no way to log this."

AI Trust / Verification

[40 notes]

"If it reads wrong and I don't notice, my whole month's budget is off. I need to be able to check." – P4

Manual Entry Fatigue

[42 notes]

"Entering every purchase individually just felt like a second job." – P10

Category Mismatch

[29 notes]

"Groceries doesn't fit my spending. Default categories don't match Pakistani habits (no ration, no rickshaw)."

Competitive Landscape: Identifying the Market Gap

	Khata Notebook	Bank Apps	Generic OCR	Cashew / Wallet	DailyKhata (Proposed)
Receipt Support	None ✗	None ✗	Printed only	Basic OCR	Handwritten + Mixed Urdu/English ✓
Input Mode	Manual	Digital only	Camera	Manual	Voice (NLP) + Camera + Manual ✓
Categorization	User-defined	N/A ✗	N/A ✗	Generic Western	Local defaults + Custom ✓
Error Correction	N/A ✗	N/A ✗	Auto-commit	Auto-commit	Inline edit-before-save ✓
Accessibility	High	Low ✗	Low ✗	Low ✗	Offline-first, icon-forward onboarding ✓

Target Personas: Designing for Two Extremes

Rukhsana Bibi (Ammi Jaan)

Age: 43 | Homemaker | Digital Literacy: Low

Current State:
Manages Rs. 40,000 budget via physical Khata; keeps faded receipts in a kitchen drawer.

Pain Point:
Apps feel 'too English' and complicated. Distrusts automation.

Key Scenario (The Camera Flow):
Wants to snap a handwritten bazaar receipt, easily verify the AI read it correctly in Urdu, and save it instantly without navigating complex menus.

Zaid Malik (The Hustle Student)

Age: 22 | CS Student/Freelancer | Digital Literacy: High

Current State:
Uses a custom Notion table after abandoning rigid budget apps like Wallet.

Pain Point:
Categorization takes too long; massive workflow friction for small daily expenses.

Key Scenario (The Voice Flow):
Pays a rickshaw Rs.200 cash, speaks into the app to log the expense instantly without typing, then later reviews granular data in a desktop-like 'Pro View'.

Implications for Design: From Research to Blueprint

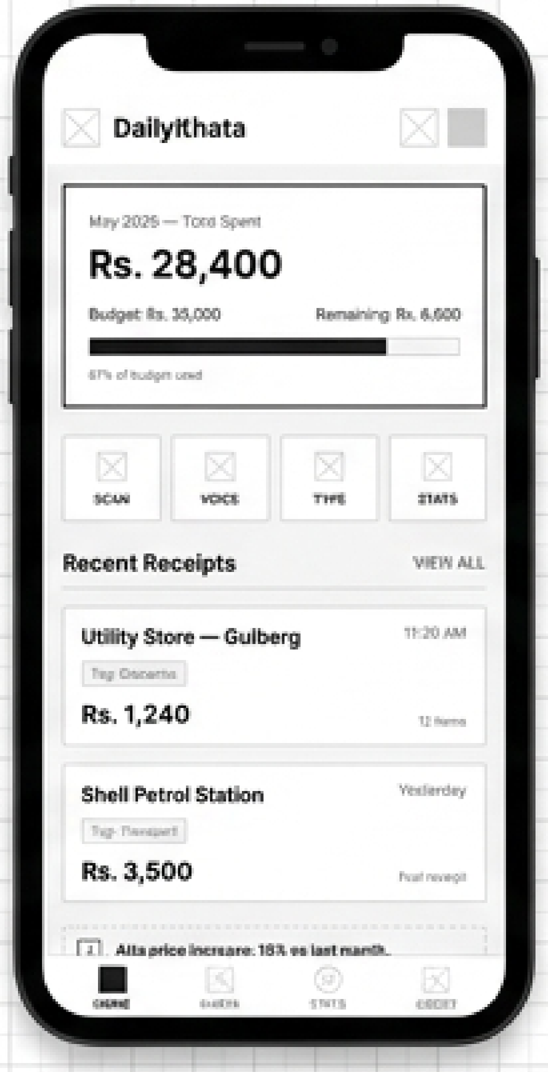
Constraint / Need (M02)	UX Implication	DailyKhata Feature (M03)
Handwritten Urdu/English receipts are the norm + Users actively distrust AI accuracy.	Auto-commit is dangerous. Radical transparency and user control are required.	Inline edit-before-save verification with yellow “uncertainty” highlighting.
High volume of no-receipt cash transactions (e.g., rickshaws, street stalls).	Need a zero-friction logging mechanism completely independent of paper or typing.	Voice-First natural language logging (“Spent 200 on rickshaw”).
Unreliable internet & low digital literacy in Group A users.	App must work instantly without account creation or cloud reliance to prevent drop-off.	Offline-first architecture with a 60-second, icon-heavy frictionless onboarding.

SECTION 2

Milestone 3: Initial Prototype Pcccrone & User Test Plans

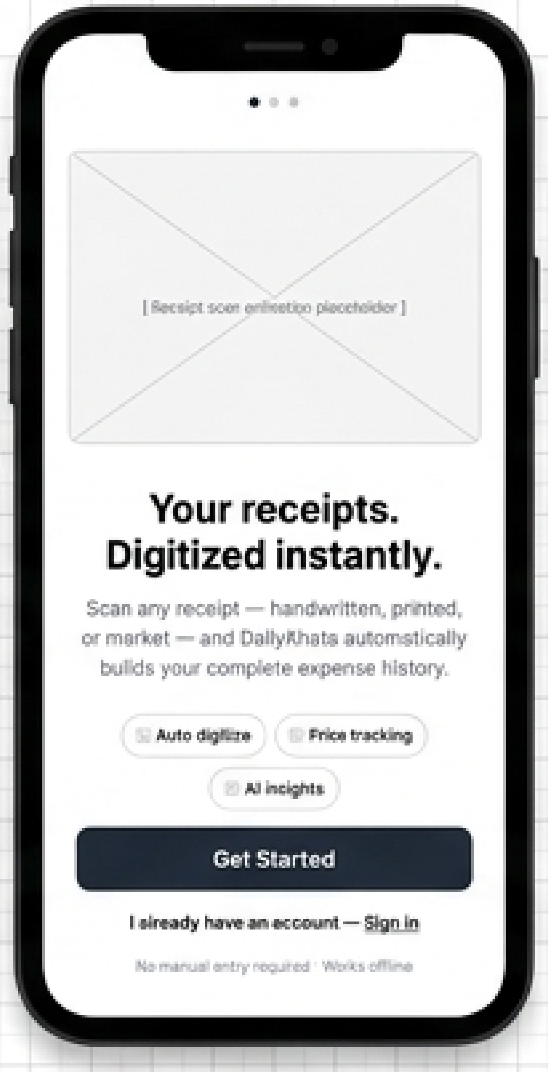
Translating validated user needs into a low-fidelity, task-based architectural blueprint.

The DailyKhata Architecture: Low-Fidelity Blueprint



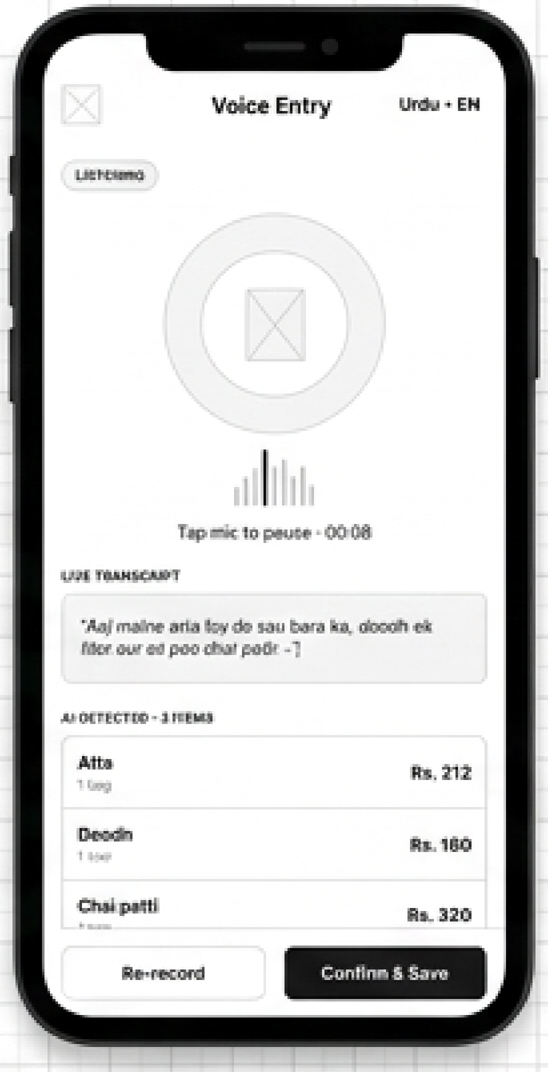
WF-01: Home Dashboard

Features minimal quick-action cards (Scan, Voice, Type) and a real-time budget progress bar. Focuses on immediate logging access.



WF-12: Onboarding

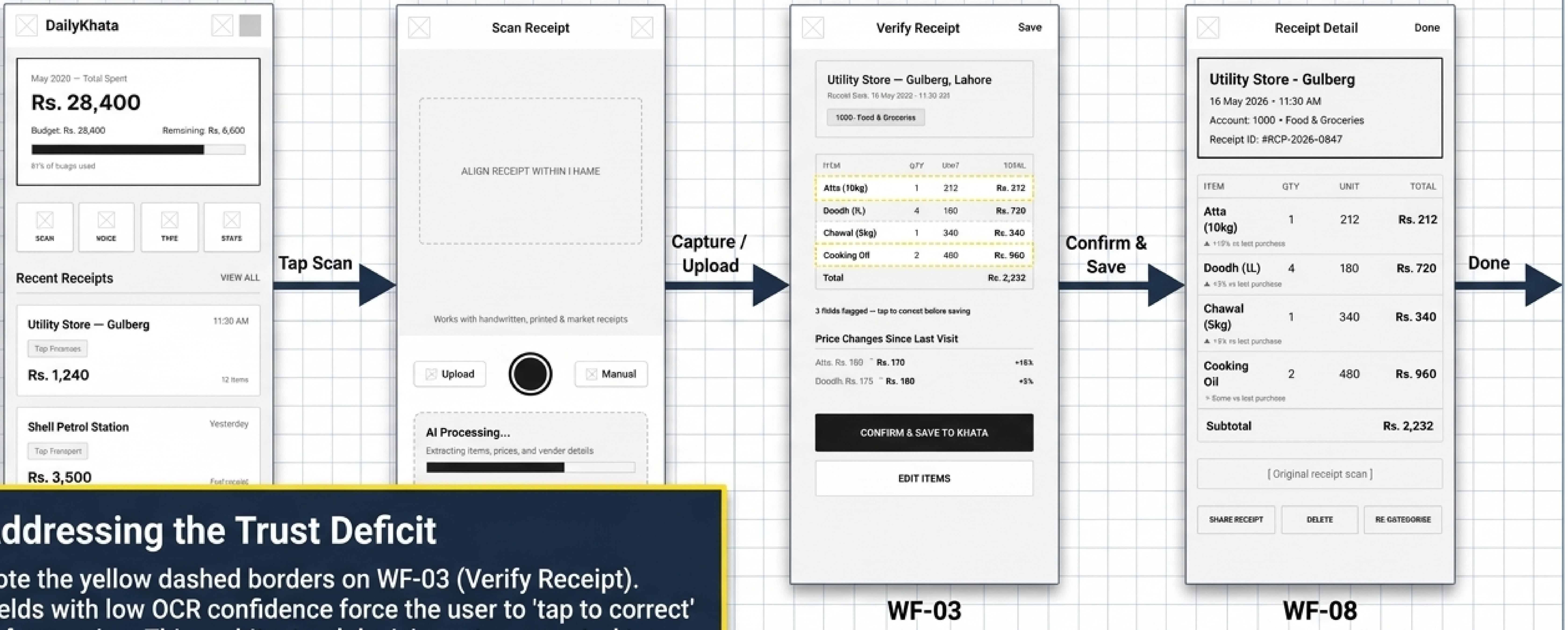
Highlighting the 'no account required' frictionless entry designed specifically to prevent drop-off among low-literacy users.



WF-13: Voice Entry

A live transcript box and AI-detected item list translate spoken Urdu/English natural language into structured data instantly.

Core Flow A: Camera-to-AI with Inline Verification



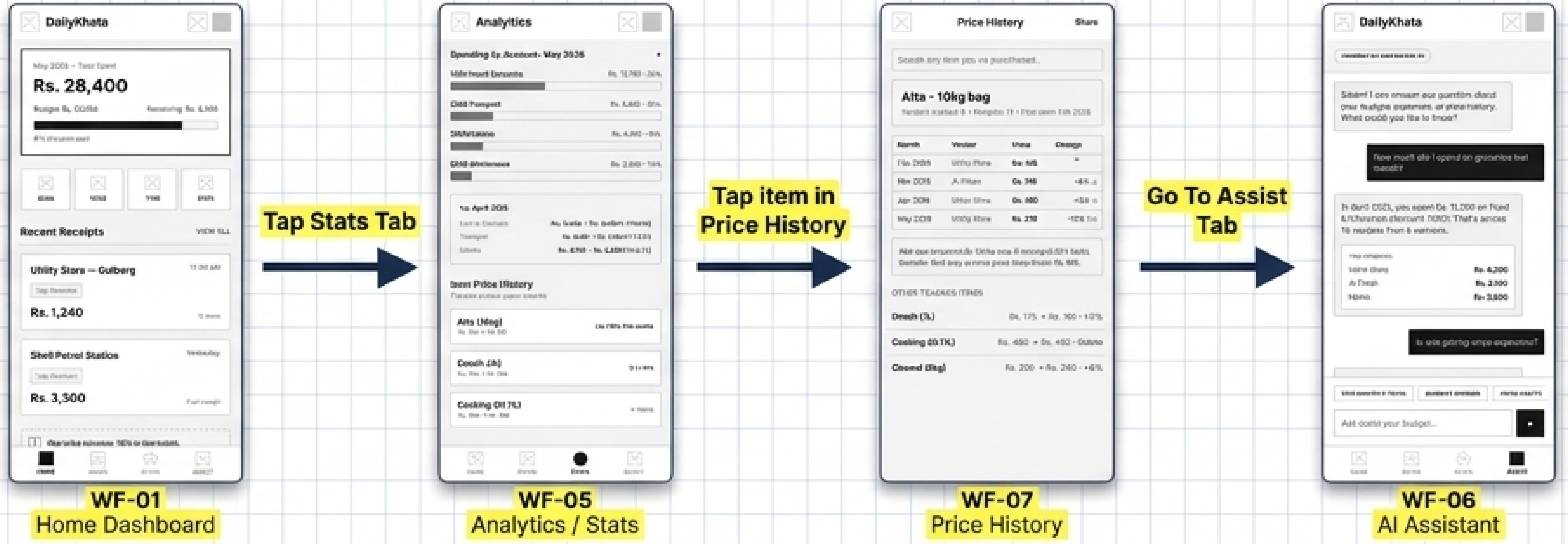
Addressing the Trust Deficit

Note the yellow dashed borders on WF-03 (Verify Receipt). Fields with low OCR confidence force the user to 'tap to correct' before saving. This architectural decision returns control to the user, preventing silent AI errors and directly answering the trust concerns identified in research.

Core Flows B & E: Analytics & The AI Assistant

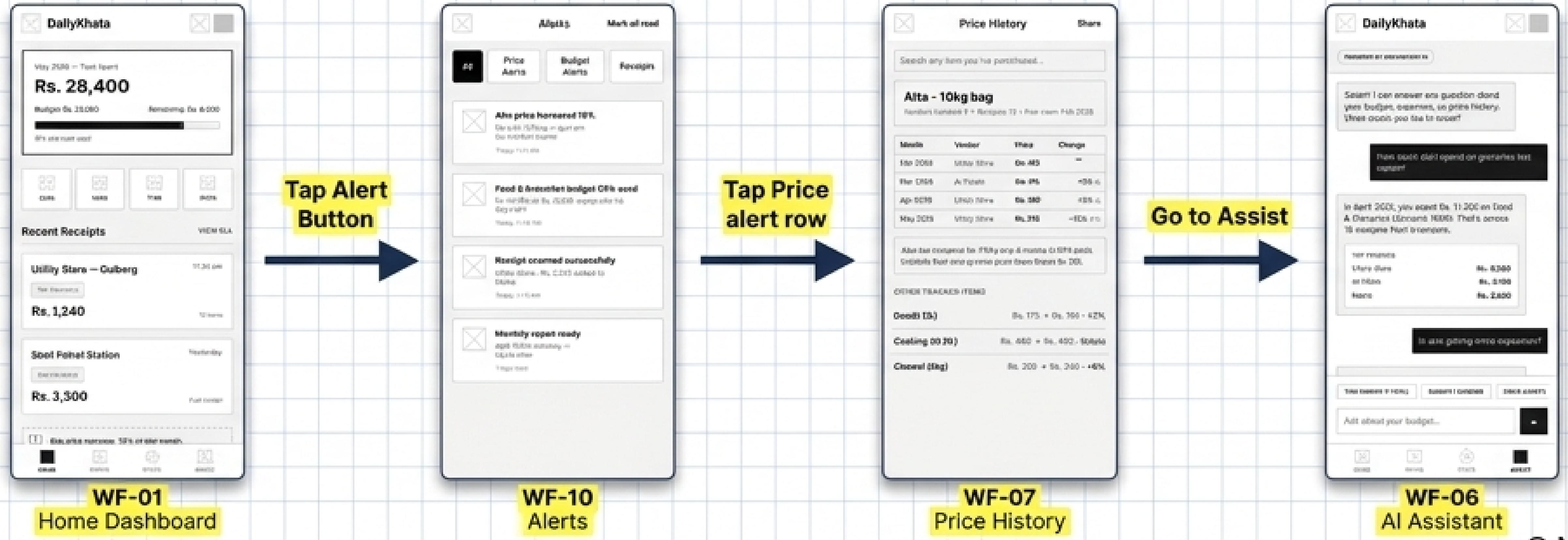
Price History (Flow B)

Users can track month-over-month inflation of specific items (e.g., Atta) across their scanned receipts, establishing the app's value beyond basic logging.



Conversational AI (Flow E)

Users bypass complex charts by asking natural language questions ("How much did I spend on groceries?"). The AI responds with plain-text summaries and inline data cards, catering to novice users.



User Test Plan: Evaluation Methodology

Minutes 0-5: Welcome & Informed Consent

Minutes 5-7: Pre-test background questionnaire

Minutes 7-10: Think-Aloud Training
(Practice task using device calculator)

Minutes 10-40: Task-based prototype walkthrough
(6 core tasks tested against specific personas)

Minutes 40-50: Post-test SUS questionnaire & debrief

Recruiting Criteria

- 5 Participants Total
- Mix of Group A (Urdu-primary, manual trackers) and Group B (Bilingual, digital drop-outs).
- Must actively manage a budget and receive physical receipts.

Task Scenarios Framework

Task #	Objective	Target Persona	Screen Flow
T1	Scan & verify a grocery receipt	Rukhsana (Homemaker)	Flow A
T2	Check Atta price history (3 months)	Rukhsana (Homemaker)	Flow B
T3	Add a cash expense manually (plumber)	Rukhsana (Homemaker)	Flow C
T4	Ask AI about last month's grocery spend	Zaid (Student)	Flow B variant
T5	Review May analytics vs April	Zaid (Student)	Flow D
T6	Act on a Cooking Oil price alert	Zaid (Student)	Flow E

Evaluation Protocols & Ethics

The Logging Sheet (Observation)

Metrics Tracked:

- Task Success (Y/N/P)
- Time to Completion (mm:ss)
- Errors / Wrong Paths taken
- Critical Incidents observed

Moderator Rules:

- No hints unless user is stuck for >90 seconds.
- Use neutral prompts only: "What would you try next?"
- Maximum 3 minutes allowed per task.

Informed Consent Framework

Ethics & Privacy:

- Voluntary participation with the right to withdraw at any time.
- Fully anonymized data (e.g., participants labeled 'P01').
- Screen and audio recording permitted strictly for internal research analysis.
- No externally shared identifiable information.

Post-Test Evaluation (SUS) & Conclusion

System Usability Scale (SUS)

10-Item SUS Questionnaire

- 10-item Likert scale (1=Strongly Disagree, 5=Strongly Agree).
- **Calculation Mechanism:**
  - Odd items (rating - 1)
  - Even items (5 - rating)
  - Sum multiplied by 2.5.
- **Result:** A 0-100 benchmark usability score.
- **Target Score:** >68 (Industry Average)

Research-to-Design Evolution

Accomplishments: Successfully mapped the chaotic reality of Pakistani informal retail into a structured, dual-interface digital prototype.

The Core Pivot: Shifted from assuming generic OCR was sufficient to realizing a realizing a hybrid input model with mandatory AI verification was required to build user trust.

Next Steps: Execute testing plan, analyze SUS scores, and iterate into high-fidelity design.